

Bucket Sort

- Works on floating-point numbers between range 0.0 to 1.0
- Inputs should be uniformly and independently distributed across $[0,1]$ to get a running time of $O(n)$

But work is done very well, this is also fast.



Selection

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Eraser

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0.79, 0.13, 0.64, 0.39, 0.20, 0.89, 0.53, 0.42, 0.06, 0.94

0
1
2
3
4
5
6
7
8
9

BUCKET-SORT (A)

- 1) let $B[0 \dots n - 1]$ be a new array
- 2) $n = A.length$
- 3) for $i = 0$ to $n - 1$
- 4) make $B[i]$ an empty list
- 5) for $i = 1$ to n
- 6) insert $A[i]$ into list $B[\lfloor nA[i] \rfloor]$
- 7) for $i = 0$ to $n - 1$
- 8) sort list $B[i]$ with insertion sort
- 9) concatenate the lists together
 $B[0], B[1], \dots, B[n - 1]$ (in order)

Second, what we have here,



¹0.79, ²0.13, ³0.64, 0.39, 0.20, 0.89, 0.53, 0.42, 0.06, 0.94



0.79 x 10

[7.9]

0.13 x 10

[1.3] = 1

$O(1) \times n$
 $O(n)$

BUCKET-SORT (A)

1) let $B[0 \dots n-1]$ be a new array

2) $n = A.length$

3) for $i = 0$ to $n-1$

4) make $B[i]$ an empty list

→ 5) for $i = 1$ to n

6) insert $A[i]$ into list $B[\underbrace{\lfloor nA[i] \rfloor}_{7}]$

7) for $i = 0$ to $n-1$

8) sort list $B[i]$ with insertion sort

9) concatenate the lists together
 $B[0], B[1], \dots, B[n-1]$ (in order)

I hope all the points have been cleared.



0.¹79, 0.²13, 0.³64, 0.39, 0.20, 0.89, 0.53, 0.42, 0.06, 0.94

0	0.06	79 X 10
1	0.13	9
2	0.20	7
3	0.39	3

3 } = 1

$O(1) \times 4$
 $O(4)$

BUCKET-SORT (A)

- 1) let $B[0 \dots n-1]$ be a new array
- 2) $n = A.length$
- 3) for $i = 0$ to $n-1$
- 4) make $B[i]$ an empty list
- 5) for $i = 1$ to n 7
- 6) insert $A[i]$ into list $B[\lfloor nA[i] \rfloor]$
- 7) for $i = 0$ to $n-1$
- 8) sort list $B[i]$ with insertion sort
- 9) concatenate the lists together
 $B[0], B[1], \dots, B[n-1]$ (in order)

For i is equal to 0 to N-1.



0.¹79, 0.²13, 0.³64, 0.39, 0.20, 0.89, 0.53, 0.42, 0.06, 0.94

0	0.06
1	0.13
2	0.20
3	0.39
4	0.42
5	0.53
6	0.64
7	0.79
8	0.89
9	0.94

0.79 x 10

[7.9]

0.13 x 10 7

[1.3] = 1

$O(1) \times n$
 $O(n)$

BUCKET-SORT (A)

1) let $B[0 \dots n-1]$ be a new array

2) $n = A.length$

3) for $i = 0$ to $n-1$

4) make $B[i]$ an empty list

→ 5) for $i = 1$ to n

6) insert $A[i]$ into list $B[\underbrace{\lfloor nA[i] \rfloor}_{7}]$

7) for $i = 0$ to $n-1$

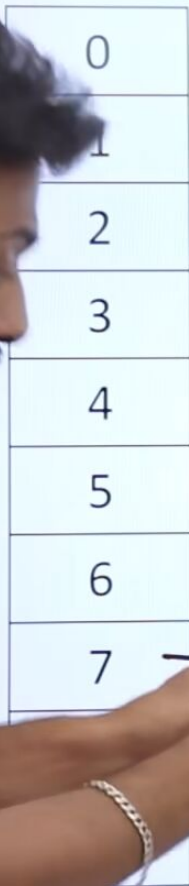
8) sort list $B[i]$ with insertion sort

9) concatenate the lists together
 $B[0], B[1], \dots, B[n-1]$ (in order)

Here you have every bucket.



If the elements are like this: 0.79, 0.78, 0.795, 0.74,.....



0
1
2
3
4
5
6
7

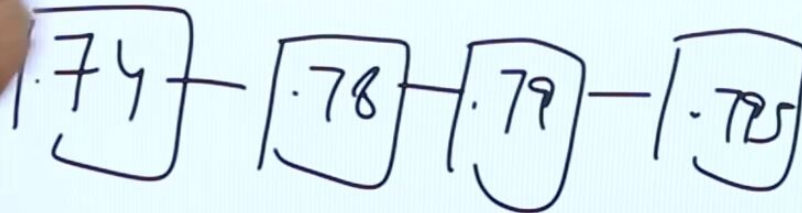


Means you have to insert the insertion sort internally here.

If the elements are like this: 0.79, 0.78, 0.795, 0.74,.....

0
1
2
3
4
5
6
7
8
9

$O(n)$

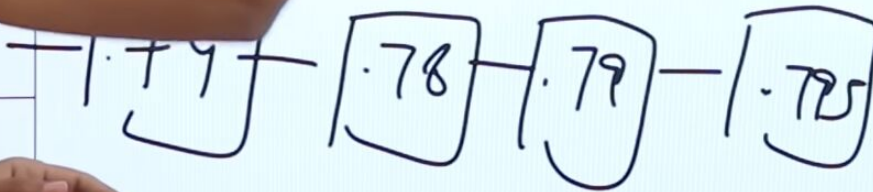


Then order of n time will take your insertion sort.

If the elements are like this: 0.79, 0.78, 0.795, 0.74,.....

0
1
2
3
4

$O(n) \times n$



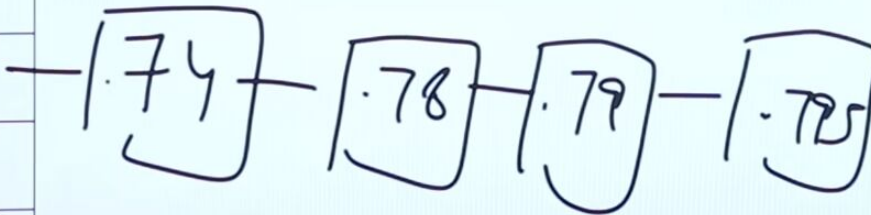
So in worst case, your order of n square can go.

If the elements are like this: 0.79, 0.78, 0.795, 0.74,.....

0
1
2
3
4
5
6
7
8
9

$$\underline{O(n^2)}$$

$$O(n) \times n$$



Okay?